

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE & NAME : CSA16 Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Q1. Dataset: Hospital Patient Segmentation

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

SSE Values

K	SSE
1	600
2	350
3	220
4	200
5	190

Question (BL4 – 8 Marks):

A healthcare organization uses K-Means clustering to group patients based on their medical expenses and health risk scores. Interpret the elbow point from the SSE values provided and determine the optimal number of clusters. Discuss how selecting this number of clusters affects patient segmentation and healthcare decision-making.

Q2. Dataset: E-Commerce Order Analysis

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000

O5	Clothing	27000
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Question (BL4 – 8 Marks):

An online retailer applies hierarchical clustering to group customer orders based on product category and order value. A dendrogram is generated from the dataset. Interpret how the clusters are formed and identify the appropriate number of clusters. Explain the significance of the selected clusters in understanding customer purchasing behaviour.

Q3. Dataset: Mobile App User Behaviour

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Question (BL5 – 8 Marks):

A mobile application company uses the HAC algorithm to segment users based on their activity patterns. Interpret how users are assigned to clusters using probability-based membership values. Discuss the advantages of soft clustering over hard clustering for customer behavior analysis.

Q4. Dataset: Agricultural Crop Monitoring

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Question (BL4 – 8 Marks):

An agricultural research center performs clustering on farm data before and after applying normalization. Interpret the differences observed in the clustering results. Explain how normalization influences cluster formation and improves the reliability of agricultural data analysis.

Q5. Dataset: Employee Performance Evaluation

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Silhouette Scores

Model	Score
K=2	0.48
K=3	0.68
K=4	0.52

Question (BL5 – 8 Marks):

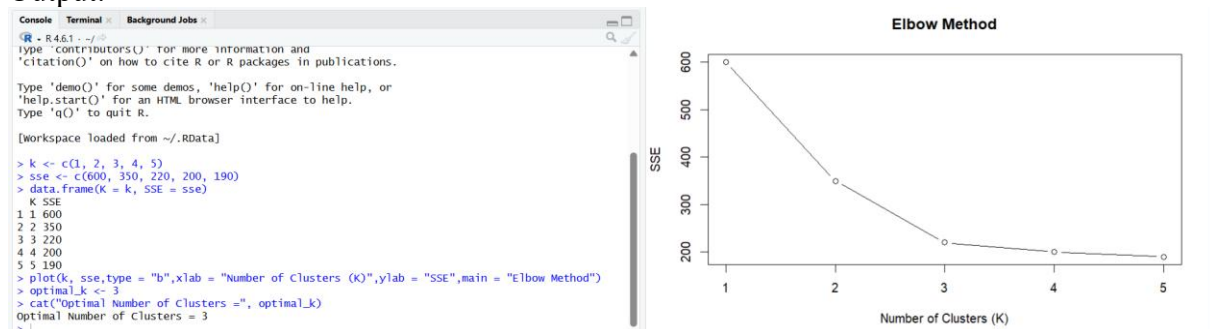
A company clusters employees based on productivity, attendance, and skill assessment scores. Analyse the silhouette scores obtained for different clustering models and determine the most suitable number of clusters. Justify your answer and explain how the selected model contributes to effective employee performance evaluation.

Answers:

1) Code:

```
k <- c(1, 2, 3, 4, 5)
sse <- c(600, 350, 220, 200, 190)
data.frame(K = k, SSE = sse)
plot(k, sse, type = "b", xlab = "Number of Clusters (K)", ylab = "SSE", main = "Elbow Method")
optimal_k <- 3
cat("Optimal Number of Clusters =", optimal_k)
```

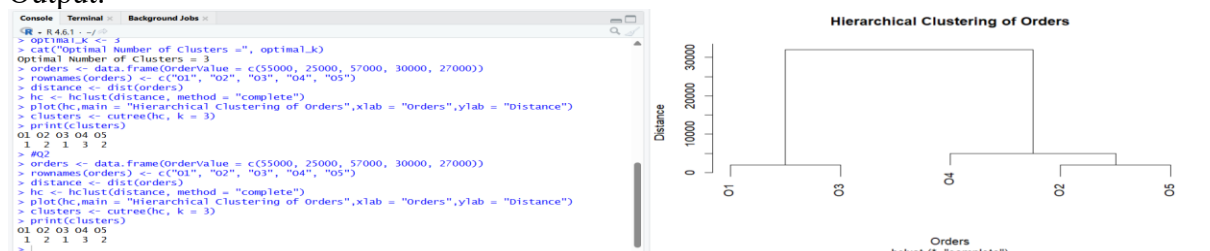
Output:



2) Code:

```
orders <- data.frame(OrderValue = c(55000, 25000, 57000, 30000, 27000))
rownames(orders) <- c("O1", "O2", "O3", "O4", "O5")
distance <- dist(orders)
hc <- hclust(distance, method = "complete")
plot(hc, main = "Hierarchical Clustering of Orders", xlab = "Orders", ylab = "Distance")
clusters <- cutree(hc, k = 3)
print(clusters)
```

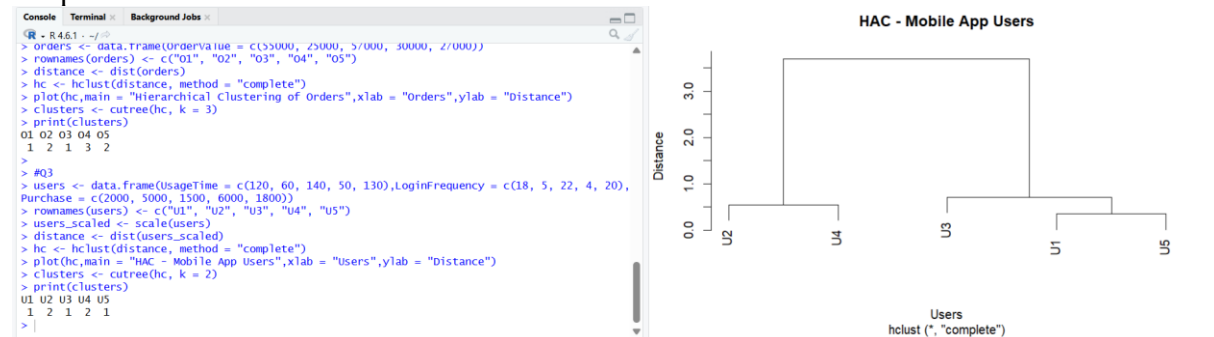
Output:



3) Code:

```
users <- data.frame(UsageTime = c(120, 60, 140, 50, 130),LoginFrequency = c(18, 5, 22, 4,
20),Purchase = c(2000, 5000, 1500, 6000, 1800))
rownames(users) <- c("U1", "U2", "U3", "U4", "U5")
users_scaled <- scale(users)
distance <- dist(users_scaled)
hc <- hclust(distance, method = "complete")
plot(hc,main = "HAC - Mobile App Users",xlab = "Users",ylab = "Distance")
clusters <- cutree(hc, k = 2)
print(clusters)
```

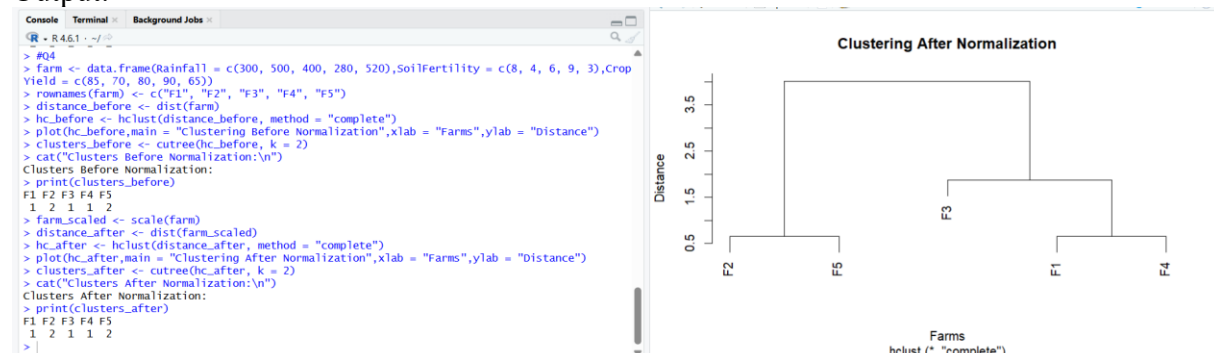
Output:



4) Code:

```
farm <- data.frame(Rainfall = c(300, 500, 400, 280, 520),SoilFertility = c(8, 4, 6, 9,
3),CropYield = c(85, 70, 80, 90, 65))
rownames(farm) <- c("F1", "F2", "F3", "F4", "F5")
distance_before <- dist(farm)
hc_before <- hclust(distance_before, method = "complete")
plot(hc_before,main = "Clustering Before Normalization",xlab = "Farms",ylab = "Distance")
clusters_before <- cutree(hc_before, k = 2)
cat("Clusters Before Normalization:\n")
print(clusters_before)
farm_scaled <- scale(farm)
distance_after <- dist(farm_scaled)
hc_after <- hclust(distance_after, method = "complete")
plot(hc_after,main = "Clustering After Normalization",xlab = "Farms",ylab = "Distance")
clusters_after <- cutree(hc_after, k = 2)
cat("Clusters After Normalization:\n")
print(clusters_after)
```

Output:



5) Code:

```
k <- c(2, 3, 4)
```

```
silhouette_score <- c(0.48, 0.68, 0.52)
```

```
data.frame(K = k,Silhouette_Score = silhouette_score)
```

```
best_k <- k[which.max(silhouette_score)]
```

```
cat("Best Number of Clusters =", best_k, "\n")
```

```
cat("Highest Silhouette Score =", max(silhouette_score))
```

```
plot(k, silhouette_score,type = "b",pch = 19,xlab = "Number of Clusters (K)",ylab =  
"Silhouette Score",main = "Silhouette Score Analysis")
```

Output:

